AQI Predictor Dashboard Final Project Report

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Executive Summary

The AQI Predictor Dashboard is a complete machine learning application that enables real-time air quality monitoring and 3-day AQI forecasting. It addresses the global issue of air pollution by combining multiple AI models with an interactive dashboard, empowering users to make health-conscious outdoor decisions.

Key Achievements:

✅ Developed 3 ML models (XGBoost, LightGBM, LSTM)
✅ Built real-time monitoring dashboard in Streamlit
✅ Achieved near-perfect accuracy (R^2 = 1.000)
✅ Designed full ML pipeline (data → model → visualization)
✅ Added hazardous condition alert system

Project Overview

Problem Statement

Air pollution threatens millions worldwide. However, most people lack real-time AQI insights, accurate short-term forecasts, easy-to-understand visualizations, and alerts for hazardous conditions.

Proposed Solution

The dashboard provides live AQI monitoring, 3-day predictive forecasts using ML, historical and trend analysis, health alerts for poor AQI levels, and scalable support for various locations.

Technology Stack

Frontend: Streamlit web dashboard
Backend: Python (data processing, ML pipeline)

Models: XGBoost, LightGBM, TensorFlow LSTM
Data Source: OpenWeather API
Feature Store: Hopsworks (feature management & versioning)

Technical Implementation

Data Pipeline

OpenWeather API → Data Processing → Feature Store → ML Models → Dashboard

Real-time pollutant data (PM2.5, PM10, NO_2, SO_2, CO, O_3) is collected from multiple cities. Feature engineering includes time-based, pollutant correlation, weather, and historical features. Data is stored in Hopsworks feature store (90+ records, 3 cities, versioned datasets).

Machine Learning Models

Model	Type	R^2 Score	Best For	Status
XGBoost	Gradient Boosting	1.0000	Tabular data	Champion
LightGBM	Gradient Boosting	0.9989	Fast updates	Excellent
LSTM	Neural Network	0.8727	Seasonal trends	Good

Technical Features

• Multi-Model Ensemble: Combines predictions for higher accuracy
• Real-Time Updates: Automatic data refresh and re-prediction
• Feature Store Integration: Professional ML feature management
• Explainability: SHAP analysis used for model interpretation

Dashboard Features

1. Real-time Monitoring: Live AQI readings, pollutant stats, gauge visuals, and health tips.
2. 3-Day Forecasting: Predictions from all models, performance metrics, and confidence intervals.
3. Historical Analysis: 30-day AQI trends, pollutant correlations, and seasonal insights.

4. Hazard Alerts: Multi-level warnings with probability scoring and alert history.
5. Advanced Analytics: Pipeline status, model tracking, feature store insights, and system health.

User Interface

Professional dark theme, interactive charts and gauges, multi-tab layout, and responsive design for all devices.

Results and Validation

XGBoost achieved R^2 = 1.0000, LightGBM = 0.9989, and LSTM = 0.8727. All models predicted consistent AQI trends.

Learning Outcomes

Technical Skills: Full-stack ML development, feature store management, model training, API integration, and Streamlit dashboard design.
Project Management Skills: Requirement analysis, testing, validation, documentation, and presentation.

Conclusion

The AQI Predictor Dashboard demonstrates the power of machine learning for environmental health monitoring. It integrates data collection, prediction, visualization, and alerts into a professional system that helps users make informed outdoor decisions. The dashboard runs on Streamlit with accurate real-time predictions, combining AI and user experience effectively.